

Differential Calculus — Chain Rule

Chapter 2: Quotient of Nested Radical Function | Page 33

PROBLEM STATEMENT

Radical Quotient & Chain Rule

Find the derivative with respect to x of the function:

$$y = \frac{\sqrt{1 - \sqrt{x}}}{\sqrt{1 + \sqrt{x}}} = \sqrt{\frac{1 - \sqrt{x}}{1 + \sqrt{x}}}$$

STEP 1 প্রদত্ত করণীয়ুক্ত ভগ্নাংশটিকে অখণ্ড ঘাত আকারে প্রকাশ করি (Expressing as Single Fractional Power)

$$y = \left(\frac{1 - \sqrt{x}}{1 + \sqrt{x}} \right)^{1/2}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{dy}{dx} = \frac{1}{2} \left(\frac{1 - \sqrt{x}}{1 + \sqrt{x}} \right)^{-1/2} \cdot \frac{d}{dx} \left(\frac{1 - \sqrt{x}}{1 + \sqrt{x}} \right) = \frac{1}{2} \sqrt{\frac{1 + \sqrt{x}}{1 - \sqrt{x}}} \cdot \frac{d}{dx} \left(\frac{1 - \sqrt{x}}{1 + \sqrt{x}} \right)$$

STEP 3 অভ্যন্তরীণ মূলদ ভগ্নাংশের অন্তরজে ভাগ বিধি প্রয়োগ করি (Applying Quotient Rule to Inner Expression)

$$\frac{d}{dx} \left(\frac{1 - \sqrt{x}}{1 + \sqrt{x}} \right) = \frac{(1 + \sqrt{x}) \left(-\frac{1}{2\sqrt{x}} \right) - (1 - \sqrt{x}) \left(\frac{1}{2\sqrt{x}} \right)}{(1 + \sqrt{x})^2}$$

STEP 4 সাধারণ গুণনীয়ক $-\frac{1}{2\sqrt{x}}$ কমন নিয়ে লব সরলীকরণ করি (Simplifying Intermediate Quotient Derivative)

$$= \frac{-\frac{1}{2\sqrt{x}} [(1 + \sqrt{x}) + (1 - \sqrt{x})]}{(1 + \sqrt{x})^2} = \frac{-\frac{1}{2\sqrt{x}} (2)}{(1 + \sqrt{x})^2} = -\frac{1}{\sqrt{x}(1 + \sqrt{x})^2}$$

STEP 5 উভয় অংশ গুণ করে অনুবন্ধী রাশির বর্গের পার্থক্যের সূত্রে সাজাই (Combining Factors via Conjugate Identity)

$$\frac{dy}{dx} = \frac{1}{2} \frac{\sqrt{1 + \sqrt{x}}}{\sqrt{1 - \sqrt{x}}} \cdot \left(-\frac{1}{\sqrt{x}(1 + \sqrt{x})^2} \right) = -\frac{1}{2\sqrt{x}(1 + \sqrt{x})\sqrt{(1 - \sqrt{x})(1 + \sqrt{x})}} = -\frac{1}{2\sqrt{x}(1 + \sqrt{x})\sqrt{1 - x}}$$

FINAL ANSWER $\frac{dy}{dx} = -\frac{1}{2\sqrt{x}(1 + \sqrt{x})\sqrt{1 - x}}$

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- **Radical Power Rule:** $\frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}} \frac{du}{dx}$.
- **Quotient Derivative of Linear Radicals:** For $\frac{1-t}{1+t}$, derivative is $\frac{-2}{(1+t)^2}$. With $t = \sqrt{x}$, multiply by $\frac{1}{2\sqrt{x}}$.
- **Conjugate Difference of Squares:**

$$\sqrt{1-\sqrt{x}} \cdot \sqrt{1+\sqrt{x}} = \sqrt{(1-\sqrt{x})(1+\sqrt{x})} = \sqrt{1-x}$$

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Sign Error in Quotient Rule:** Forgetting that $\frac{d}{dx}(1-\sqrt{x}) = -\frac{1}{2\sqrt{x}}$ carries an essential minus sign.
- **Incomplete Algebraic Simplification:** Leaving the final answer as $-\frac{1}{2\sqrt{x}(1-\sqrt{x})^{1/2}(1+\sqrt{x})^{3/2}}$ without grouping $(1-\sqrt{x})(1+\sqrt{x}) = 1-x$.
- **Radical Division Confusion:** $\frac{\sqrt{1+\sqrt{x}}}{(1+\sqrt{x})^2} = \frac{1}{(1+\sqrt{x})^{3/2}} = \frac{1}{(1+\sqrt{x})\sqrt{1+\sqrt{x}}}$.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Inner Radical Realness:** $x \geq 0$.
- **Outer Radical Realness:** $\frac{1-\sqrt{x}}{1+\sqrt{x}} > 0 \implies 1-\sqrt{x} > 0 \implies \sqrt{x} < 1 \implies x < 1$.
- **Non-zero Denominators in Derivative:** $x \neq 0$ and $x \neq 1$.
- **Strict Differentiability Domain:** Domain = $(0, 1)$.

BOX 4: TRIGONOMETRIC SUBSTITUTION ALTERNATIVE ($X = \cos^2 2\theta$)

- Let $\sqrt{x} = \cos 2\theta$, where $0 < \theta < \frac{\pi}{4}$:

$$y = \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} = \sqrt{\frac{2\sin^2 \theta}{2\cos^2 \theta}} = \tan \theta$$

- Since $2\theta = \arccos(\sqrt{x}) \implies \theta = \frac{1}{2} \arccos(\sqrt{x})$:

$$\frac{dy}{dx} = \sec^2 \theta \cdot \frac{d\theta}{dx} = (1 + \tan^2 \theta) \cdot \frac{1}{2} \left(-\frac{1}{\sqrt{1-(\sqrt{x})^2}} \right) \cdot \frac{1}{2\sqrt{x}}$$

$$\frac{dy}{dx} = \left(1 + \frac{1-\sqrt{x}}{1+\sqrt{x}} \right) \cdot \left(-\frac{1}{4\sqrt{x}\sqrt{1-x}} \right) = \left(\frac{2}{1+\sqrt{x}} \right) \left(-\frac{1}{4\sqrt{x}\sqrt{1-x}} \right) = -\frac{1}{2\sqrt{x}(1+\sqrt{x})\sqrt{1-x}}$$

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- **Step 1-2 (Power Representation & Chain Setup):** 1.0 Mark
- **Step 3 (Quotient Rule Differentiation):** 1.0 Mark
- **Step 4 (Numerator Simplification):** 1.0 Mark
- **Step 5 (Conjugate Factoring & Final Answer):** 1.0 Mark
- **Total Allocation:** 4.0 Marks